

# Algebra II with Trigonometry

## Chapter 6.4

### Olympiad Solutions

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## Solutions

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**1. Let  $f(x) = \tan(2x) + \tan(2x + \pi) + \tan(2x + 2\pi)$ . Find the value of  $f(\frac{\pi}{8})$ .**

**Solution:**

Recall that  $\tan(\theta + n\pi) = \tan \theta$  for any integer  $n$ .

- $\tan(2x + \pi) = \tan(2x)$
- $\tan(2x + 2\pi) = \tan(2x)$

So,

$$f(x) = \tan(2x) + \tan(2x) + \tan(2x) = 3 \tan(2x)$$

Now evaluate at  $x = \frac{\pi}{8}$ :

$$f\left(\frac{\pi}{8}\right) = 3 \tan\left(2 \cdot \frac{\pi}{8}\right) = 3 \tan\left(\frac{\pi}{4}\right) = 3 \times 1 = \boxed{3}$$

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**2. Identify all real  $x$  in  $[0, 2\pi)$  such that  $\cot x = \sec x$ .**

**Solution:**

Given:

$$\cot x = \sec x \implies \frac{\cos x}{\sin x} = \frac{1}{\cos x} \implies \cos^2 x = \sin x$$

Let  $y = \sin x$ . Then  $\cos^2 x = 1 - y^2$ , so

$$1 - y^2 = y \implies y^2 + y - 1 = 0 \implies y = \frac{-1 \pm \sqrt{5}}{2}$$

But  $y = \sin x \implies -1 \leq y \leq 1$ .

- $\frac{-1+\sqrt{5}}{2} \approx 0.618$  is valid.
- $\frac{-1-\sqrt{5}}{2} \approx -1.618$  is not in  $[-1, 1]$ .

So  $\sin x = \frac{-1+\sqrt{5}}{2}$ .

All  $x$  in  $[0, 2\pi)$  with this sine value are:

$$x_1 = \arcsin\left(\frac{-1 + \sqrt{5}}{2}\right)$$

$$x_2 = \pi - \arcsin\left(\frac{-1 + \sqrt{5}}{2}\right)$$

Thus, the solutions are:

$$\boxed{x = \arcsin\left(\frac{-1 + \sqrt{5}}{2}\right), \quad x = \pi - \arcsin\left(\frac{-1 + \sqrt{5}}{2}\right)}$$

**3. Let  $S$  denote the sum of all distinct  $x$  in  $(0, 2\pi)$  where  $\csc x = 2$ . Find  $S$ .**

**Solution:**

$\csc x = 2 \implies \sin x = \frac{1}{2}$ . In  $(0, 2\pi)$ ,  $\sin x = \frac{1}{2}$  at  $x_1 = \frac{\pi}{6}$ ,

$x_2 = \pi - \frac{\pi}{6} = \frac{5\pi}{6}$ .

Thus,

$$S = \frac{\pi}{6} + \frac{5\pi}{6} = \boxed{\pi}$$


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**4. Compute the product of all solutions  $x$  in  $(0, \pi)$  to the equation  $\tan 3x = 0$ .**

**Solution:**

$$\tan 3x = 0 \implies 3x = n\pi, n \in \mathbb{Z}$$

In  $(0, \pi)$ , possible  $n$  are  $n = 1, 2$ :

$$3x_1 = \pi \implies x_1 = \frac{\pi}{3}$$

$$3x_2 = 2\pi \implies x_2 = \frac{2\pi}{3}$$

So the product is:

$$\frac{\pi}{3} \times \frac{2\pi}{3} = \frac{2\pi^2}{9}$$

Thus,

$$\boxed{\frac{2\pi^2}{9}}$$


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**5. Given  $y = 3 \tan(4x - \pi)$ , determine the distance between two consecutive vertical asymptotes.**

**Solution:**

The vertical asymptotes of  $y = \tan(\theta)$  occur at  $\theta = (2n + 1)\frac{\pi}{2}$ .

So, for  $y = 3 \tan(4x - \pi)$ :

Set

$$4x - \pi = (2n + 1)\frac{\pi}{2} \implies 4x = (2n + 1)\frac{\pi}{2} + \pi = (2n + 1)\frac{\pi}{2} + \frac{2\pi}{2} = (2n + 1)\frac{\pi}{2} + \frac{2\pi}{2} = (2n + 2)\frac{\pi}{2}$$

$$x = \frac{2n+3}{8}\pi$$

Consecutive  $n$  differ by 1:

So, distance is

$$\left[ \frac{2(n+1)+3}{8}\pi - \frac{2n+3}{8}\pi \right] = \frac{2}{8}\pi = \frac{\pi}{4}$$

So,

$$\boxed{\frac{\pi}{4}}$$

**6. For  $y = \csc kx$  where  $k > 0$ , suppose that the graph between  $x = 0$  and  $x = \pi$  displays exactly three vertical asymptotes. Find the value of  $k$ .**

**Solution:**

Vertical asymptotes of  $y = \csc(kx)$  occur where  $\sin(kx) = 0$ , i.e.,  
 $kx = n\pi \implies x = \frac{n\pi}{k}$ .

In  $x \in [0, \pi]$ , the values of  $n$  go from  $n = 0$  ( $x = 0$ ) to  $n = k$  ( $x = \pi$ ), inclusively.

Thus, the number of asymptotes is  $n = 0, 1, \dots, k$ , which totals  $k + 1$  asymptotes in  $[0, \pi]$ .

We want exactly 3 asymptotes, i.e.,  $k + 1 = 3 \implies k = 2$ .

Thus,

$$\boxed{2}$$